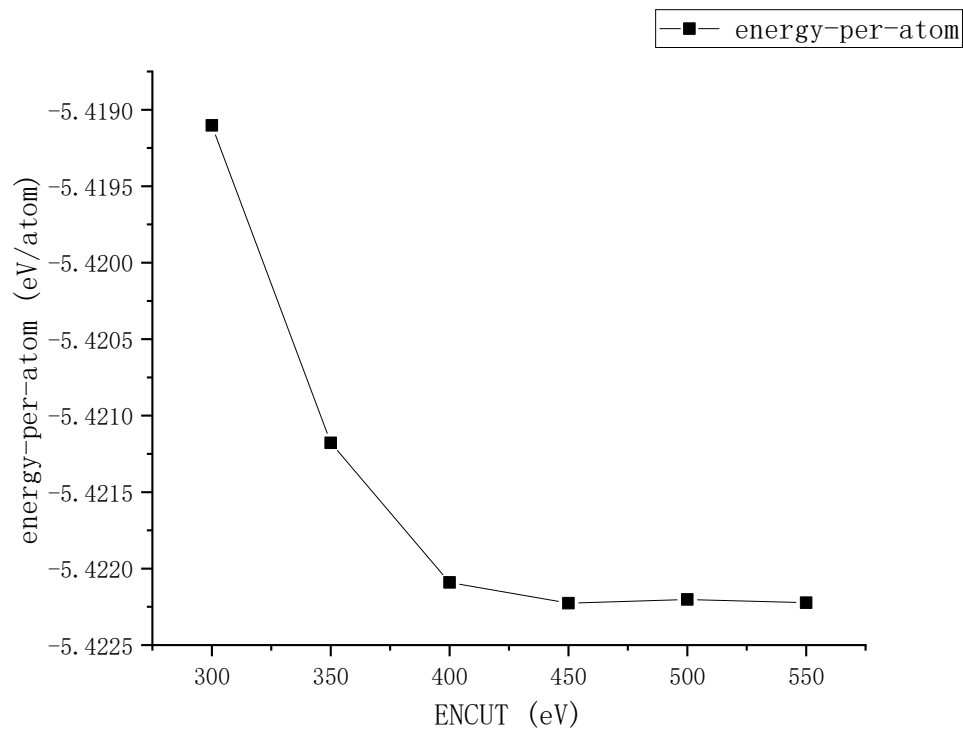


Q1



The $|\Delta E|$ changes are:

300 $\rightarrow$ 350: 2.08 meV/atom

350 $\rightarrow$ 400: 0.91 meV/atom

400 $\rightarrow$ 450: 0.14 meV/atom

450 $\rightarrow$ 500: 0.03 meV/atom

500 $\rightarrow$ 550: 0.0228 meV/atom

Using 1 meV/atom tolerance, changes become negligible by $ENCUT = 400$ eV.

Q2

(a)

Title line: a free-text comment like structure name, notes, etc.

Global scale: a single number. If 1.0, lattice vectors are interpreted as absolute lengths Å. If another number, it scales the lattice vectors.

Lattice vectors: three 3-component rows.

Element line: list of element symbols in the order you will supply coordinate counts/positions.

Counts line: number of atoms of each element in the same order.

Coordinate block: the atomic positions. First a keyword Direct or Cartesian. If Direct, coordinates are fractions with respect to the lattice vectors above. The order of coordinates must match the element ordering and counts.

(b)

```
Cu3Au-L12_a3.80
1.0
3.80000000 0.00000000 0.00000000
0.00000000 3.80000000 0.00000000
0.00000000 0.00000000 3.80000000
Au Cu
1 3
Direct
0.00000000 0.00000000 0.00000000
0.50000000 0.50000000 0.00000000
0.50000000 0.00000000 0.50000000
0.00000000 0.50000000 0.50000000
```

The element line *Au Cu* and counts *1 3* match the coordinate ordering: first the single *Au* at (0,0,0), then the three *Cu* positions. Direct is used. These fractional positions are exactly the Wyckoff coordinates. Lattice vectors are cubic with $a = 3.80$ Å and scale 1.0.

(c)

```
Cu3Au-L12_2x1x1_a3.80
1.0
7.60000000 0.00000000 0.00000000
0.00000000 3.80000000 0.00000000
0.00000000 0.00000000 3.80000000
Au Cu
2 6
Direct
0.00000000 0.00000000 0.00000000
0.50000000 0.00000000 0.00000000
0.25000000 0.50000000 0.00000000
0.75000000 0.50000000 0.00000000
0.25000000 0.00000000 0.50000000
0.75000000 0.00000000 0.50000000
0.00000000 0.50000000 0.50000000
0.50000000 0.50000000 0.50000000
```

Q3

Run1: Base

```
Cu3Au-L12_a3.80
1.0
3.800000000 0.000000000 0.000000000
0.000000000 3.800000000 0.000000000
0.000000000 0.000000000 3.800000000
Au Cu
1 3
Direct
0.000000000 0.000000000 0.000000000
0.500000000 0.500000000 0.000000000
0.500000000 0.000000000 0.500000000
0.000000000 0.500000000 0.500000000
```

Run2: Expansion

```
Cu3Au-L12_expanded_a3.876
1.0
3.876000000 0.000000000 0.000000000
0.000000000 3.876000000 0.000000000
0.000000000 0.000000000 3.876000000
Au Cu
1 3
Direct
0.000000000 0.000000000 0.000000000
0.500000000 0.500000000 0.000000000
0.500000000 0.000000000 0.500000000
0.000000000 0.500000000 0.500000000
```

Run3: Compressed

```
Cu3Au-L12_compressed_jiggled_a3.724
1.0
3.724000000 0.000000000 0.000000000
0.000000000 3.724000000 0.000000000
0.000000000 0.000000000 3.724000000
Au Cu
1 3
Cartesian
-0.027678926 0.023647121 0.017669949
1.866494148 1.834044062 0.008926568
1.881813939 -0.015974948 1.827547950
0.030712827 1.884973179 1.865622089
```

Run4: Light-shear

```
Cu3Au-L12_lightshear_a3.80
1.0
3.800000000 0.000000000 0.000000000
0.000000000 3.800000000 0.000000000
0.13269892 0.000000000 3.800000000
Au Cu
1 3
Direct
0.000000000 0.000000000 0.000000000
0.500000000 0.500000000 0.000000000
0.500000000 0.000000000 0.500000000
0.000000000 0.500000000 0.500000000
```

Table

Run	a_init(Å)	a_final(Å)	Cubic?	TOTEN(eV)	E/atom(eV)
run1	3.8000	3.5926	True	-17.096253	-4.274063
run2	3.8760	3.6258	True	-17.613251	-4.403313
run3	3.7240	3.5526	True	-16.366362	-4.091590
run4	3.8000	3.5922	False	-17.073078	-4.268469

Runs 1–3 all relaxed back to nearly the same cubic lattice parameter (~ 3.59 – 3.63 Å), even though they started with different lattice constants, which are expanded, compressed, and jiggle. This shows that the energy minimization drives the structure toward the same equilibrium geometry.

Run 4 sheared did not fully return to cubic symmetry, suggesting either incomplete relaxation of cell shape or slightly different constraints. However, its final energy and lattice constant are still very close to the others, so it's effectively converging toward the same minimum.

The total energies differ by only a few tens of $meV/atom$, which is well within numerical tolerance for VASP relaxations of similar cells.

Q4

(a)

```

AuCu_po/AB3_cf4_z21_a_c.Ab params=-1 SG=221 [ANRL doi: 10.1016/j.commatsci.2017.01.017 (part 1), doi: 10.1016/j.commatsci.2018.10.043 (part 2)]
-54.230100
 1.000000000000 0.000000000000 0.000000000000
 0.000000000000 1.000000000000 0.000000000000
 0.000000000000 0.000000000000 1.000000000000
1 3
Direct(4) [A1B3]
 0.000000000000 0.000000000000 0.000000000000 Au
 0.000000000000 0.500000000000 0.500000000000 Cu_pv
 0.500000000000 0.000000000000 0.500000000000 Cu_pv
 0.500000000000 0.500000000000 0.000000000000 Cu_pv
(UNSP_POTCAR_MODE_EXPLICIT)STOP
(AFLOW)
(UNSP_POTCAR_MODE_IMPLICIT)
(UNSP_POTCAR_FILE)Au

```

Same part: Structure type, crystal system, atom per cell, coordinate system, atom arrangement.

The lattice constant differs for AFLOW outputs normalized lattice vectors and mine specified real-length vectors.

(b)

```

AFLOW label      : AB3C_oC20_63_a_cf_c
params           : a,b/a,c/a,y2,y3,y4,z4
params values    : 3.1003,3.27443,2.48086,0.077,0.747,0.631,0.936

```

```

1.000000
 1.55015000000000 -5.07585766450000 0.00000000000000
 1.55015000000000 5.07585766450000 0.00000000000000
 0.00000000000000 0.00000000000000 7.69141025800000
2 6 2
Direct(10) [A2B6C2]
 0.00000000000000 0.00000000000000 0.00000000000000 Mg
 0.00000000000000 0.00000000000000 0.50000000000000 Mg
 0.92300000000000 0.07700000000000 0.25000000000000 O
 0.07700000000000 0.92300000000000 0.75000000000000 O
 0.36900000000000 0.63100000000000 0.93600000000000 O
 0.63100000000000 0.36900000000000 0.43600000000000 O
 0.36900000000000 0.63100000000000 0.56400000000000 O
 0.63100000000000 0.36900000000000 0.06400000000000 O
 0.25300000000000 0.74700000000000 0.25000000000000 Si
 0.74700000000000 0.25300000000000 0.75000000000000 Si

```

Its space group is *Cmcm* and crystal system is orthorhombic.

(c)

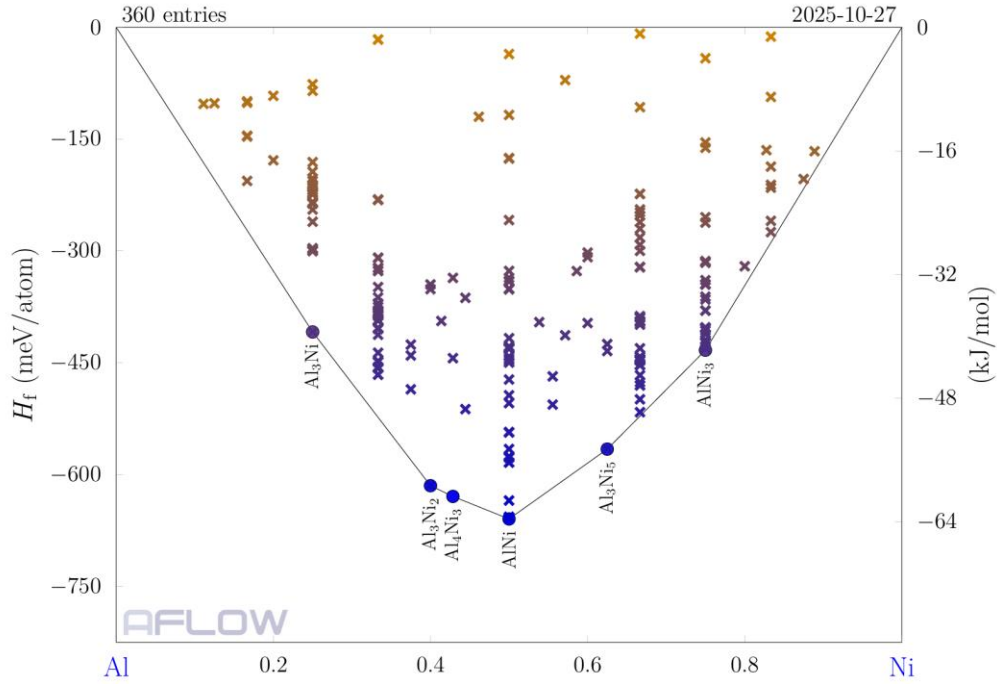
Ca is larger than *Mg* and *Ge* is larger than *Si*. For a more realistic guessed structure it might need scale the lattice parameter *a* up and keep *b/a* and *c/a* the same. So, $a_{scale} = 1.15a \approx 3.5653$

1.000000			
1.78265000000000	-5.83716263950000	0.00000000000000	
1.78265000000000	5.83716263950000	0.00000000000000	
0.00000000000000	0.00000000000000	8.84501015000000	
2 2 6			
Direct(10) [A2B2C6]			
0.00000000000000	0.00000000000000	0.00000000000000	Ca
0.00000000000000	0.00000000000000	0.50000000000000	Ca
0.25300000000000	0.74700000000000	0.25000000000000	Ge
0.74700000000000	0.25300000000000	0.75000000000000	Ge
0.92300000000000	0.07700000000000	0.25000000000000	O
0.07700000000000	0.92300000000000	0.75000000000000	O
0.36900000000000	0.63100000000000	0.93600000000000	O
0.63100000000000	0.36900000000000	0.43600000000000	O
0.36900000000000	0.63100000000000	0.56400000000000	O
0.63100000000000	0.36900000000000	0.06400000000000	O

Its space group is *Cmcm* and crystal system is orthorhombic.

Structurally it can substitute any element labels into the prototype to produce candidate structures. But need to balance the charge and radius of ions for Mg^{2+} Al^{3+} Nb^{5+} have difference charge and radius.

Q5



(a)

Stable: Al_3Ni ; Al_3Ni_2 ; Al_4Ni_3 ; $AlNi$; Al_3Ni_5 ; $AlNi_3$.

Metastable: Al_8Ni $\Delta H = 78.942$ meV/atom

$AlNi_7$ $\Delta H = 13.014$ meV/atom

(b)

Hypothetical structure:

Composition: $AlNi$

Formation energy: -480 meV/atom

The tie-line at 50% composition is lower than -480 meV/atom. The hull minima in that region sit is -659.517 meV/atom in the figure. So $AlNi$ at -480 meV/atom would sit above the convex hull and the vertical distance to the hull $\Delta H = 179.517$ meV/atom. So, it is clearly metastable.

A realistic physical reason is competition among lower-energy ordered intermetallic. In the $Al - Ni$ system, multiple ordered compounds, like Al_3Ni ; Al_3Ni_2 ; Al_4Ni_3 , already achieve very low formation energies. These have specific atomic arrangements and symmetries that optimize bonding between Al and Ni atoms. Therefore, it sits above the hull, which means thermodynamically unstable and decompose into a mixture of the more stable neighboring phases.

Q6

(a)

```
ten510-hju1040login01 Q61$ afLOW --proto=08_cf8_225_a_b:0:Mg:Ca --pocc_params=S8-0.5x8-0.5xC_S1-1x0
0MgCa/08_cf8_225_a_b.08:P0CC_P8-0.5x8-0.5xC_P1-1x0 params=-1 SG=225 1aURL doi: 10.1016/j.commatsci.2017.01.017 (part 1), doi: 10.1016/j.commatsci.2018.10.043 (p
art 2))
-40.287550 0.001 0.001
 0.000000000000 0.500000000000 0.500000000000
 0.500000000000 0.000000000000 0.500000000000
 0.500000000000 0.500000000000 0.000000000000
1=1 1=0.5 1=0.5
Direct(3) Partial [A1B0.5C0.5]
 0.000000000000 0.000000000000 0.000000000000 0 pocc=1
 0.500000000000 0.500000000000 0.500000000000 Mg pocc=0.5
 0.500000000000 0.500000000000 0.500000000000 Ca pocc=0.5
```

This creates a wurtzite-type ($Ca_{0.5}Mg_{0.5}$) structure with 50% Mg and 50% Ca mixing on the anion sublattice A-sites, while the cation sublattice B-sites is fully O , representing a single-sublattice 50/50 disordered.

Q7

